Introduction to for loops

Have you ever tried to withdraw money from your bank's ATM, but keyed in your password incorrectly? As you probably experienced, you only have a certain number of tries before the system will lock you out. Sure, it's frustrating if you're just typing too quickly or temporarily forget your PIN, but it's a great safeguard that helps protect your bank accounts. Well, the software in many cash machines runs using a for loop. This detects how many attempts have been made on a single transaction before blocking the user. And in this video, we're going to learn how that works. A FOR loop iterates over a sequence of values.

A simple example of a for loop is to iterate over a sequence of numbers. For x in range five... print x. Notice how the structure of a for loop is similar to the syntax of a typical Python statement. The first line indicates the distinguishing keyword. In this case, for. And, like functions and other expressions that start a distinct code block, it ends with a colon. The body of the for loop is indented to the right.

What's different in this case is that we have the keyword in. Also, between the keywords for and in, we have the name of a variable, x. This variable will take each of the values in the sequence the loop iterates through. In this example, x takes the values zero, one, two, three, and four. We don't have to use x. We could use any term we want: n, number, monkey. It doesn't matter, as long as we maintain consistency between what we name it here and how we refer to it below.

Ok, now let's examine the range function. The range function is a python function that returns a sequence of numbers starting from zero, increments by one by default, and stops before the given number. It can be used in while or for loops. In Python and many other programming languages, a range of numbers will start with the value zero by default. The list of numbers generated will be one less than the given value. Check it out: By default, our range function starts with zero. After the first iteration, the value will be one, the second iteration outputs two, and so forth.

Whatever code we put in the body of the loop will be executed on each of the values, one value at a time. You can also use a for loop to read in a file and iterate over the file line by line. The "with open" statement uses the file path to read in the file. In this case, it's a text file containing The Zen of Python, a famous poem written by software engineer Tim Peters in 1999. For easier notation, assign it the value of f. Otherwise, we'd have to write the file path again. In the next line, start the for loop for each line. Inside, you indent, and on the next line, tell the computer to print each line. After the loop is complete, tell the computer to print, "I'm done." I once had to locate all the unique words in a 2D array of text. I used a FOR loop inside of another for loop. The outer loop ran over each column, and the inner loop iterated over each cell for the column. That's just one example. Let's check in on Magali and Fido for another.

Magali doesn't want Fido to eat too many treats, so she only gives him five treats per day. Fido wags his tail each time he gets a treat. Once he has all five, he stops wagging his tail. Good dog, Fido. Data professionals use for loops in Python and other programming languages all the time. They're one of the fundamental tools of coding. They're great for performing a repeated procedure on an object with a fixed length to create something new.

Next, we'll explore more ways to use loops.

for loop

A piece of code that iterates over a sequence of values

range()

A Python function that returns a sequence of numbers starting from zero, increments by 1 by default, and stops before the given number

Range function

1. A range of numbers will start with the value 0 by default

2. The list of numbers generated will be one less than the given value

Question

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In Python, what type of loop iterates over a sequence of values?

- If loop
- For loop
- While loop
- Else loop

Correct
In Python, a for loop is a piece of code that iterates over a sequence of values.

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